Where the Problem Stands: Contextual States on Concrete Orthomodular Lattices

A hand-held mathematical exposition

How to read this document. **Theorem / Proposition / Corollary / Lemma**: proved in this programme's own work (proof given, marked where compressed). **External Theorem**: proved in the literature, cited not re-derived. **Conjecture**: precise, open, not asserted. **Heuristic**: an unproved intuition. **Falsified**: investigated and shown false, recorded so it is not retried under a new name.

Summary

Can a concrete system of yes/no questions carry a probability assignment that is consistent at every finite stage, yet has no countably-additive completion? Formally: a two-valued state s_0 on a finite piece of a concrete σ-class L that is finitely coherent (agrees with *some* global finitely additive state) but extends to *no* global σ-additive one. If L is only an *orthomodular poset* (meets not required), the answer is **yes** — Theorem 4.1 below, ZFC, no extra hypotheses. If L must be an *orthomodular lattice* (every pair has a meet), the answer is **conjectured no** (Conjecture 2.1) and open. Three obstruction results plus one external theorem all point toward "no": the obstacle is a genuinely infinitary, ω_1-scale combinatorial question (§5), and no finite method can reach it. Two precise open routes are stated in §7.

Contents

1 The setting, from scratch

We build the objects the problem is about, slowly. Nothing here requires quantum logic; only Boolean algebras, measures, and order theory. Each definition gets a one-sentence gloss, a formal statement, and a computation.

1.1 Why orthomodular lattices, informally

Classical probability assumes every pair of events is compatible: the events of an experiment form a Boolean algebra. Drop that assumption — keep only "A implies B" and "the complement of A," but stop requiring that any two events have a joint refinement — and what survives is an *orthomodular lattice*. Everything below is pure mathematics about such structures; no physics is assumed.

1.2 Orthomodular lattices

Definition 1.1 (Orthomodular lattice)**.** ***Gloss:*** *events with a well-behaved "did not happen" operation, but no distributive law.*
A bounded lattice L $(0, 1, \wedge, \vee)$ with an order-reversing involution $a \mapsto a^{\perp}$ such that

$$a \vee a^{\perp} = 1, \qquad a \wedge a^{\perp} = 0, \qquad a \leq b \Rightarrow b^{\perp} \leq a^{\perp}, \qquad a^{\perp\perp} = a,$$

and, the orthomodular law,

$$a \leq b \implies b = a \vee (b \wedge a^{\perp}).$$

L is Boolean exactly when every two elements are compatible (distributivity holds); the interesting case is when they are not.

Example 1.2 (MO_2: the smallest non-Boolean case). Six elements: $0, 1$, atoms $a, a^\perp, b, b^\perp$, with

$$a \wedge b = 0, \qquad a \vee b = 1, \qquad a \wedge b^\perp = 0, \qquad a \vee b^\perp = 1,$$

symmetrically in $a^\perp, b^\perp$. Check orthomodularity directly: the only nontrivial instance of $a \leq b \Rightarrow b = a \vee (b \wedge a^\perp)$ has $a \leq b$ forcing $a \in \{0, b\}$, so it holds vacuously or trivially — no case produces a violation. Distributivity fails at

$$a \wedge (b \vee b^\perp) = a \wedge 1 = a, \qquad \text{but} \qquad (a \wedge b) \vee (a \wedge b^\perp) = 0 \vee 0 = 0,$$

and $a \neq 0$. So MO_2 is orthomodular but not Boolean — the minimal such example. Every finite orthomodular lattice, MO_2 included, turns out to be *tame* in the sense of Definition 1.8 — Proposition 3.7 below.

Example 1.3 (The pentagon). Five atoms $p_1, \ldots, p_5$ in a cycle; p_i compatible with p_{i+1} (indices mod 5, sharing a Boolean block $\{0, p_i, p_{i+1}, p_i^\perp \wedge p_{i+1}^\perp, 1\}$), incompatible with p_{i+2}. Finite, concrete, non-Boolean, and — used again in §8 — a standard test bed precisely because finiteness forces tameness regardless of what else is true of it.

1.3 Concrete (set-representable) orthomodular lattices

Definition 1.4 (Concrete σ-class; OMP vs. OML)**.** ***Gloss:*** *subsets of a set Ω, closed under complement and countable disjoint union; a lattice exactly when every pair also has a genuine set-theoretic meet inside the family.*

Fix Ω. A concrete σ-class *$L \subseteq \mathcal{P}(\Omega)$ contains $\emptyset, \Omega$, is closed under complement $E \mapsto \Omega \setminus E$, and closed under countable unions of pairwise-disjoint members. Order is $\subseteq$. Every such L is automatically an **OMP** (orthomodular poset — pairs need not have a meet in L). If additionally every $A, B \in L$ have a $\subseteq$-greatest lower bound in L, call L an **OML** (orthomodular lattice). The OMP vs. OML distinction is the entire subject of this document.*

Concreteness matters: an abstract orthomodular lattice can be represented in exotic non-set ways, but a concrete one is tied to honest membership facts. (The projection lattice of a Hilbert space generically fails to be concrete — Kochen–Specker — but every object below is concrete by definition, so that obstruction never enters here.)

1.4 States

Definition 1.5 (Two-valued state)**.** ***Gloss:*** *a yes/no probability, additive over disjoint unions.*

For $B \subseteq L$ closed under complement, a state is $s : B \to \{0, 1\}$, $s(\Omega) = 1$, with

$$s\Big(\bigcup_i A_i\Big) = \sum_i s(A_i)$$

for every pairwise-disjoint $\{A_i\} \subseteq B$ with union in B: finite family for finitely additive*, countable family for σ-*additive*. For $\omega \in \Omega$, $\delta_\omega(E) := \mathbf{1}[\omega \in E]$ is always σ-additive — a* Dirac *state; every other state is* non-Dirac*.*

The entire problem lives in the gap between “additive over finite unions” and “additive over countable unions.”

1.5 Blocks

Definition 1.6 (Block)**.** ***Gloss:*** *the largest all-compatible piece containing a given event — an ordinary Boolean σ-algebra sitting inside L.*

A block *of L is a maximal pairwise-compatible subfamily. Every element of L lies in some block (Zorn's lemma), and each block, on its own, is a Boolean σ-algebra of subsets of Ω.*

All non-classicality lives *between* blocks, never inside one — the single fact the rest of this document is built on (made precise as Theorem 3.1).

1.6 Essential contextual states

Definition 1.7 (Essential contextual state)**.** ***Gloss:*** *a finite probability fragment consistent at the finite level, yet incompatible with* every *countably-additive completion.*

Fix a concrete σ-class L, a finite $\perp$-closed $B \subseteq L$, and a state s_0 on B. Call s_0 finitely coherent on *L if it extends to some finitely additive state on all of L. Call s_0* σ-essential *if it is finitely coherent on L but no σ-additive state on L extends it.*

The finite-coherence clause is load-bearing, not decorative — quantified precisely in Proposition 3.2: without it, a self-contradictory fragment that no state of any kind extends would trivially (and vacuously) satisfy the definition.

Definition 1.8 (Tame)**.** *L is* tame *($\Phi(L)$) if every finitely coherent finite pattern on L extends to a σ-additive state — i.e. L carries no σ-essential state.*

2 The central question, stated

Conjecture 2.1 (The central conjecture)**.** *Every concrete σ-complete orthomodular **lattice** is tame.*

OMP (Definition 1.4, no meet requirement)	**false** — witness exists, Thm. 4.1
OML (meets required)	**open** — Conjecture 2.1

2.1 A second, equivalent, face

Theorem 2.2 (Cluster reformulation)**.** *Call $\mathcal{C} \subseteq L$ finite an* f.a.-coherent cluster *if some finitely additive state on L assigns 1 to every member. Then:*

(a) (block form) every finite pattern on a block, coherent with some finitely additive state on L, extends to a σ-additive state on L;

(b) (cluster form) every f.a.-coherent cluster $\mathcal{C}$ is assigned 1 throughout by a single σ-additive state.

are equivalent.

Proof. (a)⇒(b): given $\mathcal{C}$, let μ witness its coherence; the pattern $s_0 := \mu|_{\perp\text{-closure}(\mathcal{C})}$ is a block-level finite pattern coherent with μ, so (a) extends it σ-additively, and that extension assigns 1 throughout $\mathcal{C}$ by construction. (b)⇒(a): given a coherent block pattern s_0 on B, let $\mathcal{C} := \{A \in B : s_0(A) = 1\}$; this is an f.a.-coherent cluster, so (b) gives a σ-additive state assigning it 1 throughout, and $\perp$-closure forces the same state to assign the s_0-false members 0, so it agrees with s_0 on all of B. □

Both directions use no property of L beyond $\perp$-closure — in particular, no latticehood. Machine-verified in both directions.

We do *not* present a third independent face. The tempting split "no Dirac state realizes the pattern" / "no non-Dirac state does either" is a tautology (every σ-state is one or the other), not a reformulation — its real content is Theorem 3.4 in §3.

3 What is proved: the solid ground

3.1 All non-classicality lives between blocks

Theorem 3.1 (Blocks are Boolean σ-fields)**.** *Every block M of a concrete σ-class L, restricted to itself, is a Boolean σ-algebra of subsets of Ω: closed under complement and countable disjoint union, with every such union already inside M. Consequently a countable disjoint family is tested for σ-additivity inside the single block containing it.*

Proof sketch. Maximality of M as a pairwise-compatible family forces closure under the Boolean operations restricted to M (standard maximality argument). □

What this buys. The search for a witness reduces entirely to an *assembly* question — how finite patterns on different blocks can or cannot be jointly realized — never a within-block question.

3.2 Why finite coherence is load-bearing

Proposition 3.2 (Boolean baseline)**.** *If L is (all of) a Boolean σ-algebra and s_0 is finitely coherent on L, then s_0 extends to a Dirac state on L. So a Boolean carrier is tame for free, not merely by conjecture.*

Proof. Let $\mu \supseteq s_0$ witness coherence, B_0 the finite Boolean subalgebra generated by B. Its atoms partition Ω into finitely many pieces $a_1, \ldots, a_k$; finite additivity gives

$$1 = \mu(\Omega) = \sum_{j=1}^{k} \mu(a_j),$$

each $\mu(a_j) \in \{0, 1\}$, so exactly one a_{j_0} has $\mu(a_{j_0}) = 1$ and $a_{j_0} \neq \emptyset$. Any $\omega \in a_{j_0}$ gives δ_ω agreeing with μ on B_0, since each member of B_0 either contains a_{j_0} or is disjoint from it. □

Why it matters. Drop finite coherence and a self-contradictory pattern (no global finitely additive state agrees with it at all) trivially fails to extend σ-additively — for a reason with nothing to do with countable additivity. Proposition 3.2 shows the clause is not merely excluding degenerate cases: on the *most* classical carrier, finite coherence already forces the strongest possible outcome. Any genuine witness must live where this forcing fails — non-Boolean, as §4's theorem does.

3.3 Countably generated blocks force triviality

Theorem 3.3 (Countably generated boundary extension)**.** *Let $A \subseteq L$ be a Boolean σ-subalgebra generated by countably many $(a_n)_{n<\omega}$, faithfully σ-embedded into a σ-complete OML L whose σ-states order-separate points. Then every σ-state μ on A extends to one on L.*

Proof. Set $b_n := a_n$ if $\mu(a_n) = 1$, else $b_n := a_n^\perp$; each finite meet $b_1 \wedge \cdots \wedge b_n$ has μ-value 1. σ-additivity of μ forces

$$c_\mu := \bigwedge_{n<\omega} b_n \;\neq\; 0.$$

Order-separation in L supplies a σ-state on L charging the image of c_μ; it agrees with μ on each generator a_n, hence on all of A. □

What this buys. Any witness confined to — or faithfully, separatingly embedded in — a countably generated piece always extends. A witness, if one exists on a lattice, cannot be countably generated in this sense.

3.4 The Dirac/non-Dirac partition

Theorem 3.4 (Localization to the non-Dirac core). *For finite $\perp$-closed $B \subseteq L$ and state s_0 on B: s_0 extends to no σ-state on L $\iff$*

$$\underbrace{\bigcap\{A \in B : s_0(A) = 1\} = \emptyset}_{\textit{(i) no Dirac extension}} \quad \textit{and} \quad \underbrace{\textit{no non-Dirac } \sigma\textit{-state extends } s_0.}_{\textit{(ii)}}$$

Proof. Every σ-state is Dirac or non-Dirac, mutually exclusive and jointly exhaustive, so "none extends" $\iff$ (i) and (ii). For (i): $\delta_\omega|_B = s_0$ iff ω lies in every s_0-true member of B (the s_0-false members are automatic by $\perp$-closure). □

This is a tautology — "a state is Dirac or it isn't" — not an independent reformulation. Its use: (i) is a purely finite, checkable condition that Theorem 4.1's construction shows can be arranged *independently* of (ii). All difficulty collapses onto (ii).

3.5 The necessary condition, with its exact choice-strength

Theorem 3.5 (Non-extendability ⇔ FIP, calibrated). *For s_0 on finite $\perp$-closed B, let $V := \{A \in B : s_0(A) = 1\}$. TFAE:*

- *(1) some $\omega \in \bigcap V$ (Dirac realization);*
- *(2) s_0 extends to the Boolean σ-algebra generated by B;*
- *(3) V has the* finite intersection property *(FIP): every finite subfamily of V has nonempty intersection.*

(1) ⇒ (3) *and* (2) ⇒ (3)*: plain ZF.* (3) ⇒ (2)*: exactly the Boolean Prime Ideal theorem (BPI) — every proper filter extends to an ultrafilter — and no weaker principle suffices.*

Proof. (1) ⇒ (3): ω witnesses every finite subfamily's intersection. (2) ⇒ (3): an extension is finitely additive and two-valued on the generated algebra, so finite meets of value-1 sets stay value-1, hence nonempty. (3) ⇒ (2): FIP makes V a filter base; BPI extends it to an ultrafilter on the generated algebra; read off the two-valued state. □

Corollary 3.6. *If s_0 is σ-essential, then (no Dirac realizes it, by definition) V fails FIP: some finite subfamily of V has empty intersection. This consequence is a plain ZF fact — the contrapositive of* (1) ⇔ (3) *needs no choice; only the abstract three-way equivalence sits at BPI.*

What this buys. Any witness search reduces to finite value-1 families failing FIP at some checkable finite stage. Checked directly against Theorem 4.1: its value-1 family fails FIP at

$$|V_{\mathrm{crit}}| = 3,$$

the smallest possible stage. Reverse-math reading: the *obstruction* (finite-stage FIP failure, used as a necessary condition) is choice-free; only the *positive* extension direction needs BPI and nothing stronger.

3.6 The finite obstruction results

Proposition 3.7 (No finite witness; the two-cell obstruction)**.** *No finite concrete OMP carries a σ-essential state (finite $\Rightarrow$ every finitely additive state is automatically σ-additive — nothing infinite to fail to converge on). Further: no concrete σ-complete **OML** can contain two Boolean blocks that (a) each faithfully, separatingly represent the same decreasing clopen sequence $(U_n)_{n<\omega}$ converging to a point, and (b) send that sequence to* different *points $i \neq j$ in the two blocks — because*

$$\bigwedge_{n<\omega} U_n$$

is computed once, in the ambient lattice, and cannot equal $\{i\}$ as seen from one block and (something containing) $\{j\} \neq \{i\}$ as seen from the other.

Proof. The countable meet of a fixed decreasing family is an intrinsic feature of a σ-complete lattice, computed via complementation and countable-join closure; both faithful embeddings realize the *same* ambient meet, so it cannot disagree with itself. □

What this buys. Rules out the most natural gluing construction — two Cantor-space-like blocks glued along a shared countable boundary — regardless of where the gluing sits in the structure.

Proposition 3.8 (The countably-generated route is fully closed)**.** *Combine Theorem 3.3 with the Lusin–Souslin injection theorem (countably many countably-generated quotient algebras of a standard Borel space that jointly separate points already generate the full Borel σ-algebra): no countably-generated, faithfully separating boundary structure — however many countably-generated pieces are combined — supports a witness.*

What this buys. Any surviving construction must be genuinely, unavoidably uncountably generated.

4 The one existing witness: a proved ZFC theorem

Theorem 4.1 (A σ-essential state exists, in ZFC)**.** *There is a concrete σ-complete orthomodular **poset** L on Ω, $|\Omega| = \aleph_1$, carrying a σ-essential state. Plain ZFC; no large cardinal or independence hypothesis.*

Construction. Let M have $|M| = \aleph_1$, well-ordered with every initial segment countable.

Step 1 (Ulam matrix). Build $\{C_{\alpha,n} : \alpha < \omega_1,\ n < \omega\} \subseteq \mathcal{P}(M)$ with

$$\bigsqcup_{n<\omega} C_{\alpha,n} = M \setminus (\text{countable set}), \qquad C_{\alpha,n} \cap C_{\beta,n} = \emptyset\ (\alpha \neq \beta).$$

(Each row disjoint and co-countable; each column pairwise disjoint across rows.)

Step 2 (rigidity). Let $\Omega = M \times \{1,2,3,4\}$, $D_{\alpha,n} := C_{\alpha,n} \times \{1,2,3,4\}$, and let L be generated by all singletons, the cells $D_{\alpha,n}$, and the three sets of Step 3. Claim: every non-Dirac candidate fails. If a σ-state kills every singleton (hence, by σ-additivity, every countable set), then in each row exactly one cell has value 1. The map

$$\alpha \ \longmapsto\ n(\alpha) := \text{the unique } n \text{ with } s(D_{\alpha,n}) = 1$$

sends $\omega_1 \to \omega$; since $|\omega_1| > \aleph_0$, some fiber is uncountable, so pick $\alpha \neq \beta$ with $n(\alpha) = n(\beta) = n^*$. Then $D_{\alpha,n^*} \cap D_{\beta,n^*} = \emptyset$ (column-disjointness), so additivity forces

$$s\big(D_{\alpha,n^*} \sqcup D_{\beta,n^*}\big) = s(D_{\alpha,n^*}) + s(D_{\beta,n^*}) = 2,$$

impossible for a $\{0,1\}$-valued state. Hence no non-Dirac σ-state exists: every σ-state on L is Dirac.
Step 3 (the pattern). Let

$$A := M \times \{1,2\}, \qquad B := M \times \{1,3\}, \qquad C := M \times \{2,3\}.$$

Then

$$A \cap B = M \times \{1\}, \qquad A \cap C = M \times \{2\}, \qquad B \cap C = M \times \{3\},$$

all nonempty, but

$$A \cap B \cap C = \emptyset.$$

So $V = \{A, B, C\}$ fails FIP at stage exactly 3 — matching Corollary 3.6 at the minimum possible value. Fix a non-principal ultrafilter $\mathcal{U}$ on M and define m on L by voting through $\mathcal{U}$; a short parity computation on the four fibers shows $m(A) = m(B) = m(C) = 1$ simultaneously, consistently, with no point witnessing all three. This m is a global finitely additive state, so $s_0 := m|_{\perp\text{-closure}(A,B,C)}$ is finitely coherent.

Conclusion. s_0 is finitely coherent (by m); by rigidity (Step 2) the only candidate σ-extensions are Dirac; no $\omega \in \Omega$ lies in $A \cap B \cap C = \emptyset$. So no σ-state extends s_0: it is σ-essential. ☐

Proposition 4.2 (Not a lattice)**.** *In L, $A \wedge B$ does not exist: every member of L inside $A \cap B = M \times \{1\}$ is countable (only countable sets and their complements/finite combinations lie in L's generated structure at that fiber), and among the countable subsets of the uncountable set $M \times \{1\}$ there is no largest one — given any countable $S \subsetneq M \times \{1\}$, adjoin one more point of $M \times \{1\} \setminus S$ to get a strictly larger member of L.*

Why this is the crux. Theorem 4.1 resolves the poset form completely and says nothing about the lattice form, because its carrier is not a lattice. Every later section is about this one gap.

5 The external boundary result: measure extension on operator algebras

External Theorem 5.1 (Escolano–Peralta–Villena, 2025)**.** *Let J be a JBW*-algebra with projection lattice $P(J)$, and suppose J has no direct summand of type I_2 (the spin-factor case, the operator-algebraic analogue of MO_2). Then every bounded finitely additive measure on $P(J)$ extends to a bounded linear functional on J (and, via Hahn–Banach, every Banach-space-valued one extends to a bounded operator).* Sharp: *if J has a type-I_2 summand, there is a bounded finitely additive measure on $P(J)$ that does* not *extend.*

Mechanism, in one line. Away from type I_2, every bounded finitely additive measure on $P(J)$ is automatically uniformly continuous — via projection halving, small symmetric motions, and isoclinic interpolation — and uniform continuity upgrades to linear extendability by standard functional analysis. This is a *finite*, elementary mechanism: no uncountable cardinals, ultrafilters, or countable additivity in the sense of Definition 1.5 enter.

Remark 5.2 (What it does and does not tell us)*. Does:* narrows where an analogous phenomenon could live — the symmetry-poor, type-I_2-rich regime, nowhere else.

Does not: (1) it is a statement about JBW* projection lattices, which need carry no compatible norm or Jordan structure — the concrete σ-classes of Definition 1.4 are not, in general, of this type, so the theorem neither forbids nor supplies a witness on them directly; (2) the sharp (dual) direction's own mechanism is a finite rank-counting clash inside a 3-dimensional spin factor —

$$\tfrac{1}{3} \neq \tfrac{1}{2}$$

— categorically not the ω_1-scale infinitary mechanism Conjecture 2.1 would need if false; (3) it narrows a *location*, not a proof — it supplies no argument for whether a witness exists there.

In one sentence: Theorem 5.1 narrows a target in an adjacent world; it is not a step toward a proof here.

6 Where the difficulty actually is: the honest crux

A lattice-form witness, if one exists, must satisfy all four at once:

(i) uncountably generated (Thm. 3.3, Prop. 3.8)
(ii) symmetry-poor (Thm. 5.1, heuristic pointer only)
(iii) FIP-failing at some finite stage, yet finitely coherent (Thm. 3.5)
(iv) survives lattice closure

6.1 Why (iv) is the genuine obstacle

Theorem 4.1 achieves (i)–(iii) with room to spare and provably fails (iv) (Proposition 4.2). Requiring L to be a lattice means: for *every* pair, a meet must be manufactured and adjoined consistently with the orthomodular law and all prior additivity constraints. Doing this coherently for the uncountably many pairs forced by (i) is an ω_1-level combinatorial question: can a coherent system of forced meets and joins be built across an uncountable index set without collapsing the essential pattern to a single point?

Remark 6.1 (Why finite methods cannot reach this). Every finite construction tried has produced only tame objects — forced, not unlucky, by Proposition 3.7: finiteness makes σ-additivity automatic, so no finite object can ever exhibit (iv) failing at the uncountable scale (i) demands.

6.2 The current assessment

Remark 6.2 (Where the evidence leans — assessment, not theorem). Evidence leans toward Conjecture 2.1 being **true**:

(a) the two-cell countable-meet obstruction (Prop. 3.7);

(b) the countably-generated closure (Thm. 3.3 + Prop. 3.8);

(c) every attempted uncountable "hub" construction has, on inspection, either collapsed back to the classical center or destroyed order-separation;

(d) every candidate lattice-theoretic survival mechanism has, on inspection, either needed exactly the distributivity being ruled out, or been falsified outright (§8).

If proved, Conjecture 2.1 becomes: latticehood alone forces every finitely coherent pattern to extend σ-additively — a clean structural theorem independent of any construction above.

7 The two open directions, precisely

7.1 Route (a): does the countable-meet obstruction generalize?

Conjecture 7.1 (Generalized countable-meet obstruction). *Does Proposition 3.7's two-cell mechanism forbid* every *faithful, order-separating, countably-generated hub shared between two blocks of a concrete σ-complete OML — not just the specific Cantor-space gluing?*

If proved: eliminates every countably-generated-boundary construction at once. If it fails: the surviving instance is a concrete design template for a witness attempt.

7.2 Route (b): the distributed nonseparating quotient route

Conjecture 7.2 (Distributed nonseparating quotient route)**.** *Is there a concrete σ-complete OML, built from uncountably-generated boundary data whose σ-states do* not *order-separate points across the whole structure — evading Theorem 3.3's hypothesis by distributing nonseparating quotient copies so no global σ-state assembles, while every finite and countable sub-piece stays consistent?*

If constructible as a genuine lattice: refutes Conjecture 2.1 outright. If provably impossible: the complementary half of a full proof, alongside (a).

Read together: (a) attacks the countably-generated case in full generality; (b) attacks the uncountably-generated nonseparating case (a) does not cover. A witness, if one exists, must refute (b) constructively while already escaping (a)'s mechanism.

8 What is heuristic, not yet mathematics

Falsified 8.1 (Dimensional-equivalence as a proxy for shared-boundary agreement). Tested hypothesis: the lattice-theoretic analogue of Murray–von Neumann equivalence of projections — call two atoms p, q *equivalent* if related by a dimension-preserving symmetry of the lattice — might proxy for "forced to agree under every state assigning both value 1 on a shared boundary" (the relation actually doing work in §3). **Falsified**: on the pentagon (Example 1.3) and every finite homogeneous OML,

$$p \sim q \quad \text{for } \textit{every} \text{ pair } p, q$$

— a purely structural consequence of finite homogeneity (any two atoms are joined by a chain of complementary pairs). A relation identically true on every pair cannot proxy a relation that must be selective. The two relations are also of different type: one rank-sensitive, one rank-blind. Dead; not to recur under a new name.

Falsified 8.2 (A winding/parity invariant). Built directly on Claim 8.1: a hoped-for integer invariant tracking how forced meets/joins wind around a cycle of incompatible atoms, meant to detect whether requirement (iv) could hold. Falls with its foundation — an identically-true relation carries no invariant. Independently: the two available test objects are structurally incapable of exhibiting the phenomenon at all — the pentagon is finite, hence automatically tame (Prop. 3.7); Theorem 4.1's witness is provably *not* a lattice (Prop. 4.2), hence has no forced meets for a winding invariant to be about. Two independent reasons the idea dies.

Heuristic 8.3 (A legitimate, unexplored forwarding lead). $\mathbb{Z}_2$-valued states (valued in the two-element group, not $\{0, 1\}$ read as reals) on symmetric-difference-closed orthomodular structures are a real, previously studied class in the literature — distinct in kind from the falsified ad hoc invariant of Claim 8.2, which does not appear in the literature under any name. Whether they bear on the essential-witness question is entirely open and unexplored: named here as a place to look next, not as a result or even a formulated conjecture.

Closing remark

A witness to genuinely contextual, σ-inextendable probability data is constructed and proved, unconditionally, in ZFC (Theorem 4.1) — one order-theoretic notch below the structure the field cares about (poset, not lattice). Whether latticehood alone forces every such pattern to dissolve is Conjecture 2.1, open. Three obstruction results and one external theorem in an adjacent world point toward "yes, it dissolves," but the obstruction, if it exists, is a genuinely infinitary ω_1-question (§5) that no finite method can reach even in principle. Two precise routes (§6) remain: either one's resolution, in either direction, is genuine progress.